

Ruixuan Jia

Email: jrx23@mails.tsinghua.edu.cn

Mobile: (+86) 180-5306-7958

EDUCATION

- **Tsinghua University** Beijing, China
B.Eng. in Computer Science and Technology *Sep. 2023 – Present*
Department of Computer Science and Technology, Tsinghua University
Xinya College, Tsinghua University
GPA: 3.9/4.0 — Rank: 9/173
CET-4: 548 CET-6: 532

RESEARCH INTERESTS

Interests: Large Language Models, Coding Agents, Tool Use, Proactive Agents, Diffusion Language Models

PUBLICATIONS

- **Improving Variable-Length Generation in Diffusion Language** ICML 2026 Under Review
- **Models via Length Regularization** *arXiv:2602.07546*
Second Author
 - **Contribution** Proposed a training-free dynamic length adjustment method for diffusion language models, enabling adaptive length prediction in completion and infilling scenarios and correcting confidence bias across candidate lengths via logarithmic length regularization.
 - **Results** Achieved 51.3% and 51.5% Pass@1 on HumanEval-Infilling and McEval, respectively, significantly outperforming prior diffusion language model methods.
 - **Responsibilities** Designed and conducted experiments, including baseline comparisons, ablation studies, and result analysis.
- **ProCodeBench: A Benchmark for Proactive Code Assistance in** 2026
- **Large-Scale Real-World Development Scenarios**
Co-first Author
 - **Benchmark Construction** Introduced ProCodeBench, a large-scale benchmark for proactive code assistance grounded in real-world IDE traces, built from three consecutive days of VSCode interaction data from 1,246 developers and yielding 5,492 annotated samples.
 - **Key Findings** Identified a clear simulation-to-reality gap: real-world traces are more diverse, temporally complex, and noisy than simulated traces, and the strongest LLM Chatbot baseline reaches only 13.57% Pass@1 on real-world evaluation.
 - **Training Analysis** Showed that simulated data alone does not generalize well to real-world settings, while simulated pretraining followed by real-world fine-tuning consistently improves Pass@1 over real-only training across all tested backbones.

RESEARCH EXPERIENCE

- **Yongchao Lab, College of AI, Tsinghua University** Beijing, China
Undergraduate Research Intern *Mar. 2026 – Present*
 - **open-self-improving-LLM** Built an LLM self-evolution framework that iteratively improves agent policies through proposal, coding, evaluation, and reflection. Integrated Terminal-Bench with stable runtime/evaluator glue, trace diagnostics, partial-run handling, and skill-based reflection, and benchmarked evolved agents against Claude Code and Terminus-style baselines.
- **AI Agent Lab, College of AI, Tsinghua University** Beijing, China
Undergraduate Research Intern *Sep. 2025 – Apr. 2026*
 - **Variable-Length Generation for Diffusion Large Language Models** Proposed a training-free variable-length generation method for diffusion language models, improving existing dLLM baselines on HumanEval-Infilling and McEval.
 - **Coding Agent Inference Efficiency Optimization** Worked on improving tool-use efficiency for coding agents.
 - **Proactive Agent** Developed a benchmark based on real user data to evaluate proactive prompting in code and IDE scenarios; explored an efficient training approach to address alignment challenges arising from differences between simulated and real interactions in temporal patterns and operation granularity.
- **AIoT Research Group, Institute for AI Industry Research, Tsinghua University** Beijing, China
Undergraduate Research Intern *Jun. 2025 – Aug. 2025*
 - **Embodied Brain** Focused on KV-cache optimization to improve large language model inference efficiency.

PROJECTS

- **Instant Messaging Application** Mar. 2025 – Jun. 2025
Python, React, JavaScript, TypeScript
 - **System Development** Implemented an instant messaging application with React for the frontend and Django for the backend.
 - **Frontend Lead** Led frontend design and development; coordinated frontend-backend interfaces, set up the CI/CD pipeline, and helped the project rank 4th among all student teams.
- **5th Jittor AI Challenge, Human Skeleton Generation Track** Jun. 2025 – Oct. 2025
Team Leader, Third Prize — Python, Jittor *GitHub Repository*
 - **Competition Result** Led the team to place 7th in the track and win Third Prize with a cash award of RMB 10,000; the competition attracted 3,244 registered participants overall.
 - **Model Design** Designed human skeleton prediction and skinning prediction models based on the PointTransformer point-cloud model.
 - **Method Innovation** Proposed multi-scale skeleton prediction and joint skinning-skeleton prediction methods to improve fine-grained prediction quality.
- **RISC-V Five-Stage Pipelined CPU** Nov. 2025 – Dec. 2025
System Verilog
 - **Architecture** Implemented a five-stage pipelined CPU supporting interrupts, exceptions, virtual addressing, and serial-port access via MMIO.
 - **Instruction Support** Implemented base RISC-V instructions and selected instructions from the B extension.
- **Relational Database Management System** Oct. 2025 – Dec. 2025
C++
 - **System Development** Built a relational database system covering file management, record management, system management, and query processing.
 - **Responsibilities** Took responsibility for the file management system, record management system, and B+ tree-based index management.
- **Minidecaf Compiler** Oct. 2025 – Dec. 2025
Python
 - **Compiler Development** Built a compiler for the Minidecaf subset of C, supporting functions, global variables, and other extended features.
 - **Optimization** Implemented graph-coloring register allocation.

HONORS AND AWARDS

Tsinghua University Second-Class Freshman Scholarship (RMB 20,000, disbursed over four semesters)	2023
Third Prize, 1st Xueyan Cup University Physics Competition, Track III; Second Prize in the Optics Section	2024
Excellence Award, Jinghua Cup CTF and 9th Tsinghua Cybersecurity Technology Challenge	2025
Third Prize, Human Skeleton Generation Track, 5th Jittor AI Challenge	2025

LEADERSHIP AND SERVICE

- **22nd “Qingxi Muxiao” (Alma Mater Outreach) Program, Shuimu** Nov. 2023 – Feb. 2024
Team Member
 - **Outreach Service** Contributed to online experience-sharing sessions, promotional post design, and admissions consultation for the homecoming outreach program.
- **“Shuhai Tongyou” Project, Tang Zhongying Charity Association** Mar. 2024 – Jun. 2024
Project Member
 - **Volunteer Work** Contributed to a rural children’s reading-support initiative by assisting with project organization and volunteer coordination.
- **Xinya Class 31** Sep. 2025 – Sep. 2026
Class Monitor
 - **Class Coordination** Coordinated class affairs, information communication, class meetings, and student activities for Xinya Class 31.

贾瑞轩

邮箱: jrx23@mails.tsinghua.edu.cn

电话: (+86) 180-5306-7958

教育背景

- 清华大学
 - 计算机科学与技术专业, 本科在读
 - 计算机科学与技术系, 清华大学
 - 新雅书院, 清华大学
 - GPA: 3.9/4.0 — 排名: 9/173
 - CET-4: 548 CET-6: 532

北京, 中国
2023 年9 月– 至今

研究兴趣

研究方向: 大语言模型、Coding Agents、工具使用、主动式智能体、Diffusion Language Models

论文发表

Improving Variable-Length Generation in Diffusion Language Models via Length Regularization

ICML 2026 Under Review

- Modeling via Length Regularization
 - 第二作者
 - 主要贡献 提出一种无需训练的动态长度调整方法, 用于diffusion language models 的变长生成, 在补全与infilling 场景中实现自适应长度预测, 并通过对数长度正则缓解不同候选长度间的置信度偏置。
 - 实验结果 在HumanEval-Infilling 和McEval 上分别取得51.3% 和51.5% 的Pass@1, 显著优于已有diffusion language model 方法。
 - 本人工作 负责实验设计与执行, 包括baseline 对比、消融实验和结果分析。

arXiv:2602.07546

ProCodeBench: A Benchmark for Proactive Code Assistance in Large-Scale Real-World Development Scenarios

2026

- 共同一作
 - 基准构建 提出ProCodeBench, 一个基于真实IDE 交互轨迹的大规模主动式代码辅助benchmark; 通过VSCode 插件连续采集1,246 名开发者3 天的真实开发数据, 并构建出5,492 条标注样本。
 - 关键发现 系统分析simulated data 与real-world data 间的simulation-to-reality gap, 发现真实轨迹在行为多样性、时间结构和噪声模式上更复杂; 在真实数据评测上, 最强LLM Chatbot baseline 的Pass@1 仅为13.57%。
 - 训练结论 验证了simulated data 单独训练难以泛化到真实场景, 但simulated pretraining 再结合real-world fine-tuning 能在所有测试backbone 上稳定优于仅使用真实数据训练。

科研经历

- Yongchao Lab, 清华大学人工智能学院
 - 本科科研实习生
 - open-self-improving-LLM 构建LLM 自进化框架, 使agent policy 能在proposal、coding、evaluation 和reflection 的迭代过程中持续优化; 完成Terminal-Bench 的稳定运行与评测集成, 支持trace diagnostics、partial-run handling 和基于skill 的reflection, 并将进化后的agent 与Claude Code、Terminus-style baselines 进行比较。

北京, 中国
2026 年3 月– 至今

- AI Agent Lab, 清华大学人工智能学院
 - 本科科研实习生
 - Diffusion Large Language Models 的变长生成 提出一种无需训练的变长生成方法, 提升现有dLLM baseline 在HumanEval-Infilling 和McEval 上的表现。
 - Coding Agent 推理效率优化 围绕coding agents 的工具使用效率开展研究。
 - Proactive Agent 基于真实用户数据构建benchmark, 用于评测代码与IDE 场景下的主动式提示能力; 探索高效训练方案, 以缓解仿真交互数据和真实数据在时间模式和操作粒度上的差异所带来的对齐问题。

北京, 中国
2025 年9 月– 2026 年4 月

- AIoT 课题组, 清华大学人工智能产业研究院
 - 本科科研实习生
 - Embodied Brain 研究KV-cache 优化方法, 以提升大语言模型推理效率。

北京, 中国
2025 年6 月– 2025 年8 月

项目经历

● 即时通信应用	2025 年3 月– 2025 年6 月
● Python, React, JavaScript, TypeScript	
○ 系统开发 使用React 搭建前端、使用Django 搭建后端，实现即时通信应用。	
○ 前端负责人 负责前端设计与开发，协调前后端接口，并搭建CI/CD 流水线；项目在所有学生小组中排名第4。	
● 第五届计图人工智能挑战赛人体骨骼生成赛道	2025 年6 月– 2025 年10 月
● 队长，三等奖 — Python, Jittor	GitHub Repository
○ 竞赛结果 带队获得赛道第7 名和三等奖，奖金人民币10,000 元；比赛共有3,244 名注册选手参赛。	
○ 模型设计 基于PointTransformer 点云模型设计人体骨架预测与蒙皮预测模型。	
○ 方法创新 提出多尺度骨架预测与关节蒙皮-骨架联合预测方法，提升细粒度预测质量。	
● RISC-V 五级流水线CPU	2025 年11 月– 2025 年12 月
● SystemVerilog	
○ 体系结构实现 实现支持中断、异常、虚拟地址和通过MMIO 访问串口的五级流水线CPU。	
○ 指令支持 实现RISC-V 基础指令集及B extension 的部分指令。	
● 关系型数据库管理系统	2025 年10 月– 2025 年12 月
● C++	
○ 系统开发 实现覆盖文件管理、记录管理、系统管理和查询处理的关系型数据库系统。	
○ 负责模块 负责文件管理、记录管理以及基于B+ 树的索引管理模块。	
● Minidecaf Compiler	2025 年10 月– 2025 年12 月
● Python	
○ 编译器实现 实现面向Minidecaf 这一C 语言子集的编译器，支持函数、全局变量等扩展特性。	
○ 优化 实现基于图着色的寄存器分配。	

荣誉奖项

清华大学新生二等奖学金（共20,000 元，分四学期发放）	2023
第一届学研杯大学物理竞赛III 赛道三等奖；光学部分二等奖	2024
京华杯CTF 暨第九届清华大学网络安全技术挑战赛优胜奖	2025
第五届计图人工智能挑战赛人体骨骼生成赛道三等奖	2025

学生工作与服务

● 第22 届情系母校“水木系青唐”支队	2023 年11 月– 2024 年2 月
● 队员	
○ 活动参与 参与线上经验分享、推送制作与招生咨询等工作。	
● 唐仲英爱心社“书海童游”项目	2024 年3 月– 2024 年6 月
● 项目组成员	
○ 志愿服务 参与乡村儿童阅读支持项目，协助项目组织与志愿者管理。	
● 新雅31 班	2025 年9 月– 2026 年9 月
● 班长	
○ 班级工作 负责班级事务协调、信息传达、班会组织与集体活动统筹。	